NodeMCU ESP8266: setup and board test

For Windows + Arduino IDE 2.x. Works for both the small NodeMCU (CP2102 chip) and the bigger NodeMCU v3 "LoLin" (CH340G chip).

What is in this folder

`drivers/cp2102/`	USB driver for the **small** NodeMCU (chip next to USB says CP2102)
`drivers/ch340/`	USB driver for the **bigger** NodeMCU v3 (chip says CH340G)
`esp8266_board_test/`	Arduino test sketch (prints chip info, MAC, Wi-Fi scan, blinks LED)

0. Install the USB driver (once per PC)

Look at the small chip next to the board's USB port.

- **CP2102:** unzip `drivers/cp2102/CP210x_Universal_Windows_Driver.zip`, right-click `silabser.inf` → **Show more options** → **Install**.
- **CH340 / CH340G:** run `drivers/ch340/CH341SER.EXE` → **INSTALL**.

Check: plug in the board, open **Device Manager** → **Ports (COM & LPT)**. You should see `Silicon Labs CP210x ... (COMx)` or `USB-SERIAL CH340 (COMx)` with no yellow warning sign.

Nothing appears at all? Your USB cable is probably charge-only. Try a cable that can transfer files to a phone.

1. Add the ESP8266 board package (once only)

- **File** → **Preferences** → **Settings**. Next to **Additional boards manager URLs**, click the small window icon.
- Keep any existing line (e.g. ESP32) and add this on a **new line**:
  `https://arduino.esp8266.com/stable/package_esp8266com_index.json`
- Click **OK**, then **OK** again.

2. Install it

- **Tools** → **Board** → **Boards Manager**, search **esp8266**, install **"esp8266 by ESP8266 Community"**. Takes a few minutes.

3. Open the test sketch

- **File** → **Open** → `esp8266_board_test/esp8266_board_test.ino` (from this folder).

4. Select board and port

- Board dropdown (top bar) → **Select other board and port...**
- Search **NodeMCU 1.0** and choose **NodeMCU 1.0 (ESP-12E Module)**.
- Port: the **COMx Serial Port (USB)** that appeared when you plugged the board in → **OK**.
- Untick **Show all ports** if you see extra COM ports: those belong to the PC, not your board.
- The bottom bar should now say **"NodeMCU 1.0 (ESP-12E Module) on COMx"**.

5. Upload

- **Close the Serial Monitor first** if it is open. An open monitor locks the port and causes "could not open port" errors.
- Click **Upload** (→). NodeMCU normally uploads by itself.
- If it fails with `Failed to connect`: hold the **FLASH** button, click Upload, release FLASH when `Connecting...` appears.
- Success looks like: `Chip is ESP8266EX ... Hash of data verified ... Hard resetting via RTS pin`.

6. Check the board

- **Tools → Serial Monitor**, set **9600 baud** (bottom-right dropdown), then press **RST** on the board. The sketch uses `Serial.begin(9600)`; the monitor must use the **same** number.
- The report is printed **once at startup**, so if the monitor shows nothing, press RST again.

OK Chip ID, flash size and MAC address are printed
OK `PASS: found X networks`
OK Blue LED blinks once per second and `alive, uptime ...` keeps counting

Write the **MAC address** on a piece of tape on the board (Node 1, Node 2, ...).

7. Next board

Unplug and plug in the next one. **The COM number will probably change**, especially between CP2102 and CH340 boards, so re-select the port. Then repeat steps 5–6. The board setting stays.

Troubleshooting

Problem	Fix
No COM port appears	Wrong/missing driver (step 0) or charge-only cable.
Device Manager shows a yellow **!**	Driver missing: install the one matching the USB chip (step 0).
Gibberish in Serial Monitor	Monitor baud does not match `Serial.begin(...)` in the code. Use **9600** in both (the speed printed on the back of the NodeMCU). One short line of gibberish right after RST is normal: it is the chip's boot message at 74880 baud.
Monitor is empty	Press RST; the report only prints at startup.
`could not open port`	Close the Serial Monitor, or another program using the port.
Arduino IDE stuck on the splash screen, or downloads time out **on the college network**	The college network needs its proxy: **File → Preferences → Network → Manual proxy**, host `172.31.2.3`, port `8080`. Switch back to **No proxy** at home.
ESP8266 install fails with a GitHub download timeout	The college proxy blocks GitHub file downloads. Do step 2 once on a phone hotspot (set **No proxy** while on the hotspot).